Tutorial - 06

Q) A University wants to predict whether a student will pass or fail on exam using SVM.

Input features: Study hours

x_1	x_2	x_3	x_4	class
50	30	180	90	1
28	22	100	70	-1
55	32	190	95	1
2~~4~~	20	95	68	-1

$x = [x_1 \; x_2 \; x_3 \; x_4]$

$f(x) = w^T x + b$

if $f(x) \geq 1 \rightarrow +1$

$f(x) < 1 \rightarrow -1$

Iteration: 2

$w = [0,0,0,0]$ $b=0$, $\eta = 0.01$

$f(x) = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + b$

$= 0$

update $w = w + \eta y x_1$

$= (0,0,0,0) + 0.01 \times (1)(50, 30, 180, 90)$

$w = (0.5, 0.3, 1.8, 0.9)$

Step 2 $w = (0.5, 0.3, 1.8, 0.9)$

$x_2 = (28, 22, 100, 70)$, $y_2 = -1$

$f(x) = w^T x$

$= (0.5)(28) + (0.3)22 + 1.8(100) + 0.9(70)$

$= 263.6$

$y \times f(x) = -1 \times 263.6 = -263 < 1$ Not correct.

update weights:

$w = (0.22, 0.08, 0.8, 0.2)$

$x_3 = (55, 32, 190, 97)$, $y = 1$

$w = (0.22, 0.08, 0.8, 0.2)$

$f(x) = 185.66$

$y \times f(x) = 2 \times 185.66 = 185.66 > 1$

<u>Step 4</u> $w = (0.22, 0.08, 0.8, 0.2)$

$x_4 = (25, 20, 95, 68)$

$f(x) = 96.7$

$y \times f(x) = (-1)\, 96.7$

$= -96.7 < 1$

Accuracy $\frac{2}{4} \times 100 = 50\%$

Q) A University wants to predict whether a student will pass or fail an exam using SVM

Input features: Study hours per week

Attendance %

Assignment Score

Internal Test marks

Target variable Class

(pass = 1, fail = -1)

Studyhours	Attendance	Assignment	Internal	class
14	88	80	32	1
6	60	55	22	-1
16	92	85	40	1
5	50	50	20	-1

(P)

1) Define the hyperplane eqn for linear SVM.

2) What happens when a data point violates the margin?

3) Compute the accuracy of SVM model.

4) Give predicted class labels, calculate classification accuracy for 4 iterations with weight updation.

3rd **Step-1** Linear SVM decision function.

$$f(x) = w^T x + b$$

$$f(x) = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + b$$

prediction Rule :- If $f(x) \geq 0 \rightarrow$ It belongs to positive class (+1)

If $f(x) < 0 \Rightarrow$ It belongs to negative class (-1)

$$f(x) = w^T x + b$$

Step-2 Supportive vectors cors negative hyperplane, positive hyper plane.

$$w^T x + b = -1 \quad \text{(Negative)}$$

$$w^T x + b = 1 \quad \text{(Positive)}$$

$$\text{margin} = \frac{2}{\|w\|}$$

w = root of w square.

Step-3 Train the SVM using 4 Iterations.

updation Rule :- If $y_i(w^T x) + b$

If -1 to update values

$$w = w \cdot y * \text{learning rate}.$$

Iteration :- $x_1 = 14$, $x_2 = 88$, $x_3 = 80$, $x_4 = 3$

$\eta = 0.01$, $b = 0$

$$w = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4$$

$$w = 0$$

update = $w + \eta y x_1$

$= 0 + (0.01)(1)(14)$

$= 0.14$

$w + \eta y x_3$

$= 0 + (0.01)(1)(5$

$= 0.80$

$= w + \eta y x_2$

$= 0 + (0.01)(+1)(88)$

$= 0.88$

$(0.14, 0.88, 0.80, 0.35)$

$w_1 \quad w_2 \quad w_3 \quad w_4$

update $w + \eta y x_1$

$= 0.14 + 0.01(-1)6$

$= 0.08$

$0.8 + 0.01(-1)55$

$= 0.25$

$0.08 + (0.01)(-1)(60)$

$= 0.28$

$0.32 + (0.01)(-1)(22)$

$= 0.13$

$(0.08, 0.28, 0.25, 0.13)$

$w_1 \quad w_2 \quad w_3 \quad w_4$

update $= w + \eta y x_1$

$= 0.08 + (0.01)(1)(16)$

$= 0.24$

$= 0.28 + (0.01)(1)92$

$= -0.64$

$= 0.25 + (0.01)1(55)$

$= 1.1$

$= 0.13 + (0.01)1(40)$

$= -0.27$

update $= w + \eta y x_1$

$= 0.08 + (0.01)1(5)$

$= 0.03$

$= 0.08 + (0.01)(-1)(58)$

$= -0.5$

$= 0.08 + (0.01)(1)(80)$

$= 0.5$

$= 0.08 + (0.01)(-1)(20)$

$= 0.07$

update $= w + \eta y x_1$

$= 0.08 + (0.01)(-1)5 = 0.03$

$(0.08, 0.28, 0.25, 0.13) + (0.01 . (-1)(5, 58, 50, 20))$

$= 0.03, -0.3, 0.25, -0.07$

$$\text{margin} = \frac{2}{\sqrt{(0.03)^2 + (-0.3)^2 + (-0.25)^2 + (-0.07)^2}}$$

$$= \frac{2}{\sqrt{0.0009 + 0.09 + 0.0625 + 0.0049}}$$

$$= \frac{2}{\sqrt{0.1583}} = \frac{2}{0.3978} = 5.02$$

$$\text{Accuracy} = \frac{2}{4} \times 100 = 50\%$$